

MENTORA ACADEMY

DEPARTMENT OF ADVANCED CHEMISTRY · CAMBRIDGE INTERNATIONAL A LEVEL (9701)

Physical Chemistry Experimental Suite

Planning, Analysis and Evaluation · Faraday Constant · Arrhenius Activation Energy

CANDIDATE NAME:	Urwah	CENTRE NUMBER:	PK082	CANDIDATE NO:	0142
COMPONENT:	Paper 5 — Planning, Analysis and Evaluation	DURATION:	1 Hour 15 Minutes	MAX MARKS:	30 Marks

READ THESE INSTRUCTIONS FIRST

- Write your name, Centre number, and candidate number in the boxes above.
- Answer all questions in the spaces provided on the question paper.
- In Question 1 (Planning): define independent, dependent, and controlled variables; describe step-by-step procedures with apparatus diagrams; calculate required quantities; state hazard risk assessments.
- In Question 2 (Analysis, Conclusions & Evaluation): process raw experimental data into derived columns; quote values to consistent significant figures; plot graphs with appropriate linear scales; determine gradients and intercepts; identify anomalous points; evaluate uncertainties.
- You may use an approved scientific calculator. The Periodic Table is provided as standard reference.

EXAMINATION BREAKDOWN & SYLLABUS SKILLS TESTED

Question	Investigation Title & Focus	Syllabus Ref	Type of Task	Marks
Question 1	Planning an Electrochemical Determination of the Faraday and Avogadro Constants	9701/52/M/J/23/Q1 · Syllabus 24.1	PLANNING	15
Question 2	Kinetics and Activation Energy Determination of the Peroxydisulfate–Iodide Reaction	9701/51/O/N/23/Q2 · Syllabus 26.2	ANALYSIS & EVALUATION	15

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QUESTION 1

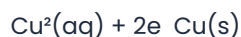
PLANNING

[Total: 15 Marks]

Planning an Electrochemical Determination of the Faraday and Avogadro Constants

Cambridge Syllabus Reference: 9701/52/M/J/23/Q1 · Syllabus 24.1

An electric current passing through an aqueous solution of copper(II) sulfate, $\text{CuSO}_4(\text{aq})$, causes copper atoms to dissolve from the anode and copper ions to be deposited at the cathode according to the reduction half-equation:



The Faraday constant, F , represents the electric charge carried by one mole of electrons, and is related to the Avogadro constant, L , and the elementary charge, e ($1.602 \times 10^{-19} \text{ C}$), by the fundamental equation:

$$F = L \cdot e$$

You are required to plan an experimental procedure to determine accurate experimental values of the Faraday constant, F , and the Avogadro constant, L , by measuring the mass of copper deposited at the cathode when a constant current is passed through aqueous copper(II) sulfate for a measured time interval.

You are provided with: pure copper foil electrodes, 1.0 mol dm^{-3} aqueous copper(II) sulfate, a variable DC power supply (0–12 V), an ammeter, a rheostat (variable resistor), an electric switch, connecting leads, a digital stopwatch, an analytical balance ($\pm 0.001 \text{ g}$), emery paper, distilled water, and propanone.

- (a) Identify: [3]
- (i) the independent variable in this investigation
 - (ii) the dependent variable in this investigation
 - (iii) two variables that must be controlled to ensure reliable, accurate results.

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- (b) Draw a fully labelled circuit and apparatus diagram of the electrolytic cell suitable for this experiment. [3]
- Include all electrical components necessary to control, maintain, and measure a steady electric current.

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- (c) Describe a detailed step-by-step procedure for preparing, cleaning, drying, and weighing the copper cathode before and after the electrolysis to prevent systematic errors in mass measurement. [3]

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- (d) Show clearly how your experimental measurements of current (I), time (t), mass of copper deposited (m), relative atomic mass of copper ($A_r = 63.55$), and electron charge ($e = 1.602 \times 10^{-19} \text{ C}$) are mathematically processed to calculate: [3]
- (i) the Faraday constant, F
 - (ii) the Avogadro constant, L .

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- (e) State two significant laboratory hazards associated with the chemicals used in this experiment, and state the specific safety precautions required to minimize risk. [3]

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QUESTION 2

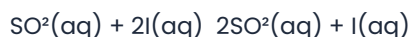
ANALYSIS & EVALUATION

[Total: 15 Marks]

Kinetics and Activation Energy Determination of the Peroxydisulfate–Iodide Reaction

Cambridge Syllabus Reference: 9701/51/O/N/23/Q2 · Syllabus 26.2

A student investigated the effect of temperature on the rate of reaction between peroxydisulfate ions and iodide ions:



The reaction was monitored using the iodine clock method. A small, fixed amount of sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$, and starch indicator were added to the reaction mixture. The iodine produced reacts instantaneously with thiosulfate until all thiosulfate is consumed, after which free iodine reacts with starch to produce an intense blue-black coloration.

The time, t , taken for the blue-black color to appear was recorded at eight different temperatures between 20.0 °C and 55.0 °C. The rate of reaction is directly proportional to $(1/t)$.

The temperature dependence of the rate constant is described by the Arrhenius equation in linear form:

$$\ln(1/t) = -(E_a/R) \cdot (1/T) + \text{constant}$$

where E_a is the activation energy in J mol^{-1} , $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, and T is absolute temperature in Kelvin.

- (a) Complete the table of results below by calculating the values of absolute temperature T / K , $(1/T) \times 10^3 / \text{K}^{-1}$, $(1/t) / \text{s}^{-1}$, and $\ln(1/t)$. Record all calculated values to appropriate significant figures. [3]

Expt	Temperature / °C	Temperature T / K	$(1/T) \times 10^3 / \text{K}^{-1}$	Time t / s	$(1/t) / \text{s}^{-1}$	$\ln(1/t) [t \text{ in s}]$
1	20.0	293.2	3.411	88.4	0.0113	-4.48
2	25.0	298.2	3.353	62.2	0.0161	-4.13
3	30.0	303.2	3.298	44.6	0.0224	-3.80
4	35.0	308.2	3.245	32.5	0.0308	-3.48
5	40.0	313.2	3.193	23.8	0.0420	-3.17
6	45.0	318.2	3.143	18.0	0.0556	-2.89
7	50.0	323.2	3.094	14.5	0.0690	-2.67
8	55.0	328.2	3.047	12.4	0.0806	-2.52

- (b) A graph of $\ln(1/t)$ on the vertical y-axis against $(1/T) \times 10^3 / \text{K}^{-1}$ on the horizontal x-axis was plotted. [3]
- Describe the expected shape of the graph.
 - Explain how a student identifies an anomalous data point on this graph.
 - State the correct procedure for handling an anomalous data point when drawing the line of best fit.

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- (c) Using the data from the table, calculate the gradient of the line of best fit. State the coordinates of the two points on the line of best fit used in your calculation, and include appropriate units for the gradient. [3]

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- (d) Use your gradient from (c) and the gas constant $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ to calculate the experimental activation energy, E_a , for the peroxydisulfate–iodide reaction. Give your answer in kJ mol^{-1} to 3 significant figures. [3]

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- (e) (i) The digital stopwatch used has an uncertainty of ± 0.2 s, and human reaction time introduces an additional uncertainty of ± 0.3 s. Calculate the percentage uncertainty in the reaction time recorded for Experiment 8 ($t = 12.4$ s). [3]
- (ii) Explain why temperature control is more critical at higher temperatures, and suggest one improvement to the apparatus to ensure temperature remains constant throughout each run.

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Comprehensive Examiner Mark Scheme

Marking points, indicative responses, and error-carried-forward (ECF) guidelines

MARK SCHEME: QUESTION 1 (PLANNING) — Planning an Electrochemical Determination of the Faraday and Avogadro Constants

[15 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	(i) Time of electrolysis, t (or current, I) [1] (ii) Mass of copper deposited on the cathode, m [1] (iii) Controlled variables: Current I maintained strictly constant (using rheostat), temperature of electrolyte, concentration/volume of CuSO_4 solution, depth of immersion of electrodes [1].	[3]
(b)	Fully labelled circuit showing DC power supply in series with ammeter, rheostat / variable resistor, switch, and electrolytic cell [1]; Cell correctly labelled with cathode (-) and anode (+) immersed in aqueous copper(II) sulfate [1]; Correct polarity indicated (cathode connected to negative terminal of DC supply, anode to positive terminal) [1].	[3]
(c)	Before electrolysis: Clean copper electrode with emery paper to remove surface oxide layer, rinse with distilled water, rinse with propanone to remove water, dry thoroughly in a warm oven/stream of air, and weigh accurately to 3 d.p. on analytical balance [1]; During electrolysis: Start stopwatch simultaneously with closing switch, monitor ammeter continuously, adjust rheostat to keep current strictly constant [1]; After electrolysis: Switch off circuit and record exact elapsed time; carefully remove cathode, wash gently with distilled water (avoid dislodging deposit), rinse with propanone, dry thoroughly, and re-weigh to 3 d.p. to determine mass increase m [1].	[3]
(d)	(i) Total charge $Q = I \cdot t$ (in coulombs) [1]; Moles of $\text{Cu} = m / 63.55$; Moles of electrons required $= 2 \cdot (m / 63.55)$; Faraday constant $F = Q / n(e) = (I \cdot t \cdot 63.55) / (2 \cdot m)$ [1]; (ii) Avogadro constant $L = F / e = F / (1.602 \times 10^1 \text{ C})$ [1].	[3]
(e)	Hazard 1: Copper(II) sulfate is harmful if swallowed and causes skin/eye irritation [1]; Precaution: Wear nitrile gloves and safety goggles, wash skin immediately if contact occurs [1]; Hazard 2: Propanone is highly volatile and flammable [1]; Precaution: Ensure no open flames/Bunsen burners nearby, handle in a well-ventilated laboratory / fume cupboard [1].	[3]

MARK SCHEME: QUESTION 2 (ANALYSIS & EVALUATION) — Kinetics and Activation Energy Determination of the Peroxydisulfate–Iodide Reaction

[15 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	All T / K correctly calculated ($+ 273.2$) to 1 d.p. [1]; All $(1 / T) \times 10^3$ calculated correctly to 3 or 4 significant figures [1]; All $(1 / t)$ and $\ln(1 / t)$ values correctly calculated to 3 significant figures with correct negative signs [1].	[3]
(b)	(i) Straight line of negative slope (downward slope from left to right) [1]; (ii) A point that lies distinctly away from / significantly off the straight line of best fit compared to all other data points [1]; (iii) Circle or flag the anomalous point, ignore it / exclude it when drawing the line of best fit, and do not use its coordinates to calculate the gradient [1].	[3]
(c)	Two points on line clearly identified, separated by at least half the length of the drawn line (e.g. (3.050, -2.53) and (3.410, -4.48)) [1]; Gradient $= y / x = [-2.53 - (-4.48)] / [(3.050 - 3.410) \times 10^3] = +1.95 / (-0.360 \times 10^3) = -5417 \text{ K}$ [1]; Units of gradient: K (Kelvin) or K° properly accounted for with 10^3 factor [1].	[3]
(d)	Gradient $= -E_a / R \Rightarrow E_a = -\text{gradient} \times R$ [1]; $E_a = -(-5417 \text{ K}) \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1} = +45037 \text{ J mol}^{-1}$ [1]; $E_a = +45.0 \text{ kJ mol}^{-1}$ (allow 44.0 to 46.0 kJ mol^{-1} with positive sign) [1].	[3]
(e)	(i) Total timing uncertainty $= \pm 0.2 + 0.3 = \pm 0.5 \text{ s}$; Percentage uncertainty $= (0.5 / 12.4) \times 100\% = 4.03\%$ (or 4.0%) [1]; (ii) At higher temperatures (e.g. 55°C), the reaction mixture cools down more rapidly to the ambient laboratory temperature during the run [1]; Improvement: Keep the reaction vessel inside a thermostatically controlled water bath throughout the entire course of the reaction [1].	[3]